

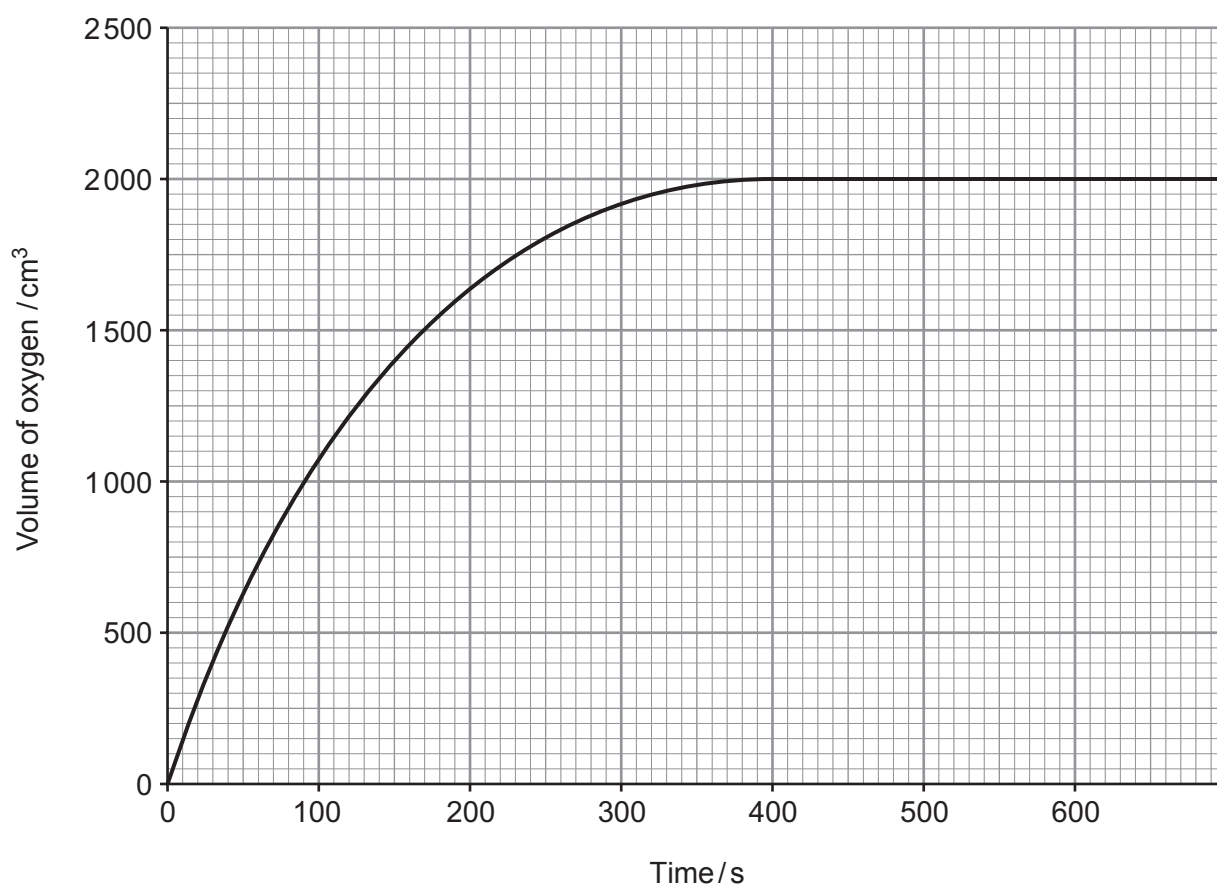
9. A student was investigating the rate of reaction for the catalysed decomposition of hydrogen peroxide to form oxygen.



The student decided to follow the reaction by measuring the volume of oxygen produced at fixed time intervals. He added a spatula measure of manganese(IV) oxide to 100 cm³ of aqueous hydrogen peroxide solution and started the timer.

- (a) Draw a labelled diagram of the apparatus he could use to carry out the experiment and measure the volume of oxygen produced. [2]

- (b) The student plotted his results to produce a graph as shown.



- (i) Use the graph to calculate the initial rate of the reaction. Show clearly, on your graph, how you obtained your answer. [2]

Initial rate = $\text{cm}^3 \text{s}^{-1}$

- (ii) Use the graph to calculate the rate of the reaction at 200 seconds. Show clearly, on your graph, how you obtained your answer. [1]

Rate = $\text{cm}^3 \text{s}^{-1}$

- (iii) Use collision theory to explain why the rates you have calculated in parts (i) and (ii) are different. [2]

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- (iv) Use the graph to calculate the concentration, in mol dm^{-3} , of the hydrogen peroxide solution used by the student. Give your answer correct to **three** significant figures. [3]

Assume that the reaction was carried out at 298 K and 1 atm pressure.

Concentration = mol dm^{-3}



- (c) A second student said that she had seen cobalt(II) chloride speed up the rate of production of oxygen very significantly in a different reaction.

Describe how the students could determine if cobalt(II) chloride is a better catalyst than manganese(IV) oxide in the decomposition of hydrogen peroxide. [3]

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